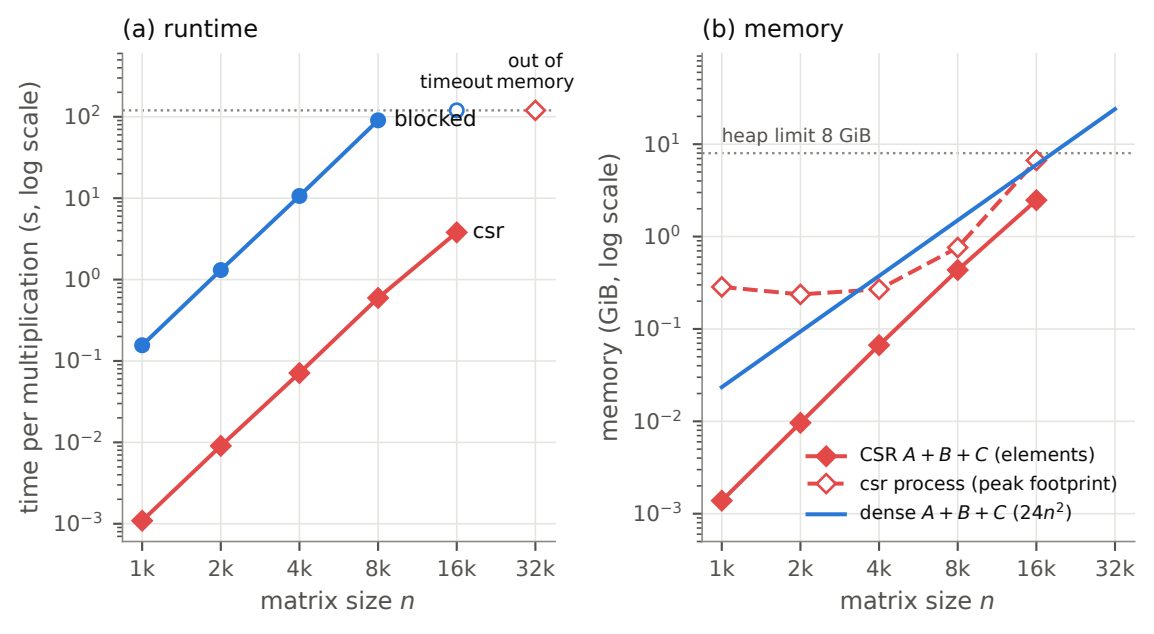
(a) runtime
out of
timeout memory
blocked
csr
time per multiplication (s, log scale)
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
1k
2k
4k
8k
16k
32k
matrix size n
(b) memory
heap limit 8 GiB
memory (GiB, log scale)
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
CSR $A + B + C$ (elements)
csr process (peak footprint)
dense $A + B + C$ ($24n^2$)
1k
2k
4k
8k
16k
32k
matrix size n